Who Pays More, and Who Can Least Afford It? Food Price Premiums, Deprivation and Abnormal Price Movements Across Selected Pakistani Urban Markets

Milestone 03 extended technical report (the submitted manuscript is the condensed M03_report.pdf)

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Notebook, code and data: github.com/Samzzh0803/Pakistan-Food-Price-Premiums-and-Deprivation (branch `milestone-03`; notebook `notebooks/M03_baseline_models.ipynb`)

Summary. Pakistan's weekly Sensitive Price Indicator (SPI) is reported as one national figure, but PBS publishes prices for 51 items in 17 cities. Milestone 02 built a 23,409-row city-item-week dataset and found that city food price premiums are negatively rank-correlated with district deprivation. Milestone 03 asks whether the same data can give **early warning** (SDG 2.c): using only what is known in week *t*, can we flag the city-item prices that will move abnormally by week *t*+1? Abnormal means a change above the item's own 90th percentile of non-zero changes, fitted on training weeks only. On 7,095 labelled rows (205 abnormal, 2.9%) we compare two naive benchmarks with logistic regression, a decision tree and KNN. We use a time-ordered split and forward-chaining cross-validation that never uses a label from the test period. On the four held-out weeks, logistic regression reaches ROC-AUC 0.692 [0.636, 0.746] (row-level bootstrap), and it is above 0.5 in each test week. We reject H3's null for these weeks, while noting that four weeks cannot establish a general result. A one-line rule that scores each item by its category's past abnormal rate does as well (0.708 [0.659, 0.753]), and it is also the cheapest alerting policy in our cost analysis. Precision stays near 5%. The models are at most a **triage aid**, not an automatic trigger. In these test weeks, alerts pay off only if a missed shock is judged to cost about 20 or more false alarms.

1. Project alignment and master dataset inventory

Team. Both members are computer scientists. One owns acquisition, parsing and provenance; the other owns statistical framing, representation, ethics and policy interpretation. **Research question.** Beneath the single national weekly SPI figure, do persistent city-level food price differences exist, are the cities paying most the ones least able to absorb it, and is the published city detail enough to detect localised price stress? **SDGs:** primary SDG 2 (Target 2.1, access to food; Target 2.c, functioning food markets and market information); secondary SDG 10 (reduced inequalities).

Changes since Milestone 01. The 7-city panel reported in Milestone 01 was a parsing error: Appendix-A stacks three tables in one sheet, and the old script read only the first. The panel covers all 17 SPI cities (8 Punjab, 4 Sindh, 2 KP, 2 Balochistan, plus Islamabad). A correction note is in the repository. Following the instructor's feedback, three external sources are now merged.

Table 1. Master dataset: 23,409 rows × 46 columns. Observational unit: 1 row = 1 item's retail price in 1 city in 1 week (17 cities × 51 items × 27 weeks; 32 of the 51 items are food).

Source	Role	Coverage	Access
PBS weekly SPI Annexure, Appendix-A	City-item prices (MIN/AVG/MAX): the panel	27 weeks, 2025-05-15 to 2026-09-24	xlsx from pbs.gov.pk release pages
PSLM District Level Survey 2019-20	District deprivation: FIES food insecurity, literacy, schooling, water	2019-20	PDF report tables (text extraction)
Digital Census 2023, Table 1	Total and urban population	2023	xlsx, one file per province
CPI (Urban), food and non-alcoholic beverages	Deflator only, not a regressor	monthly, 2025-05 to 2026-09	PDF and Monthly Review docx

2. Data ingestion, cleaning and structural readiness

2.1 Collection audit

Each PBS weekly release page (`/weekly-sensitive-price-indicator-spi-for-the-week-ended-on-DD-MM-YYYY/`) links an Annexure workbook. A rate-limited script tried the Thursday, Wednesday and Friday page for each of 73 weeks from 2025-05-15 and logged every request, URL and SHA-256 hash. It found release pages for 12 weeks, most of them weeks already retrieved for Milestone 01 (the re-downloads were byte-identical). After dropping 12 duplicate files, the panel holds 27 weeks. Other weeks may exist under URLs we did not try, so we do not treat the archive as complete. The week after our freeze date (2026-10-01) was stored, unopened, as a holdout. External files were downloaded from pbs.gov.pk and are kept unchanged with their URLs and hashes.

Validation. The parser asserts, for every week: 3 blocks, 17 cities, 51 items, 867 rows; no duplicate key; MIN ≤ AVG ≤ MAX; and zeros only as all-zero triples. Against PBS's own national columns, the cross-city minimum and maximum match PBS's national MIN and MAX in 1,377 of 1,377 item-weeks. The **geometric mean** of the city averages matches PBS's national average within 0.1% in 1,377 of 1,377. The arithmetic mean misses by a median of 0.42%. We therefore use the geometric mean as the reference price.

2.2 Reshaping and alignment

The raw sheet (179 rows × 25 columns) is not tidy. City names are column headers, each spanning MIN/AVG/MAX cells, and three tables are stacked in one sheet. We **melt** each block into one row per city × item, **concatenate** the blocks and weeks, and move PBS's national columns into a separate validation table. We then **join** the item classification (on `item_id`), the city attributes (on `city`) and the CPI (on the month of `week_end`). Every join is many-to-one, and the row count is asserted unchanged (Table 2).

Key alignment decisions.

- **Karachi:** one SPI market, but six PSLM districts and seven census districts. PSLM values are population-weighted.
- **Islamabad and Rawalpindi:** one contiguous market, kept as two administrative units.
- **District totals:** PSLM has no urban sample for Khuzdar and publishes FIES only for district totals, so all indicators use district totals. These are **deprivation measures, not income**.

Table 2. Reshaping and merge steps with keys, cardinality and the asserted row count.

operation	source	key	cardinality	rows_after
stack	3 Appendix-A blocks per workbook	same 51 item rows	rows: 7+7+3 cities x 51 = 867 per week	867
stack	27 weekly workbooks	week_end	867 x 27	23409
merge	item_mapping.csv	item_id	many-to-one (51 items)	23409
merge	city table (Census 2023 + PSLM 2019-20)	city	many-to-one (17 cities)	23409
merge	urban food CPI	calendar month of week_end	many-to-one (months)	23409

2.3 Cleaning and missing data

Nothing is deleted. Zero MIN/AVG/MAX triples (83 rows, 0.35%) are set to missing and flagged: rice IRRI-6/9 is never quoted in Gujranwala, Lahore or Sialkot, and Khuzdar has no gas quote in two weeks. PBS itself excludes these cities from its national mean. Missing counts and percentages for **every column** are in Appendix Table A1 of the Milestone 02 report. Most of the missingness is structural: weekly change columns are undefined across non-consecutive weeks (44.3% of rows), because only 15 pairs of weeks are 6-8 days apart. Table 3 gives the choice made for each kind of missingness. **Selection bias:** we exclude structural non-quotes from price comparisons, so results describe quoting markets and may still reflect selection bias, because a missing quote does not prove an item is unavailable in that city. The gaps in weekly coverage also mean the volatility analysis sees only the weeks we could retrieve.

Table 3. Kinds of missingness, the choice made for each, and why.

kind	rows	choice	reason
Structural non-quote (PBS printed 0/0/0)	83	set to missing, flag, do not impute	Imputing invents a market price PBS does not record; PBS itself excludes these cities from its national mean
Weekly change across a non-consecutive gap	10,367	leave missing	Interpolation would manufacture week-to-week changes nobody observed
Weeks not retrieved from the PBS site	–	treat as a coverage gap	No source to impute from; stated as a limitation, not filled

2.4 Transformations, scaling and encoding

Log transformation. Prices are compared in logs because premiums are multiplicative: a 10% premium means the same thing for salt and for a 20 kg flour bag. Within an item, the median skewness of prices falls from 0.22 (raw) to 0.02 (log). Prices pooled across items in PKR are not a meaningful distribution (skewness 2.13), because they mix units. The main variable is `rel_price` = log(city price / national geometric mean), which sums to zero across cities in each item-week. District urban population is strongly right-skewed (skewness 2.30, driven by Karachi). Its log (`log_pop_urban`) has skewness -0.72. **Scaling.** The four deprivation indicators are min-max scaled to 0-1 (higher = more deprived) and averaged into `deprivation_composite`. Prices are z-scored within each item and week (`price_z_item_week`). All constants come from the training weeks only. **Encoding.** Nominal strings become 0/1 dummy flags with fixed reference levels: Punjab for province, storable staple for item category (Table 4). The model-ready file `data/processed/m02/model_ready_food.parquet` (14,607 food rows × 44 columns) carries 7 dummy columns. Cities will enter H1 with sum-to-zero coding, so each coefficient is a premium relative to the average city.

Table 4. Dummy encoding example: province and item_category strings (left) become 0/1 flags (right). Reference levels: Punjab and storable_staple (all zeros).

city	province	item_category	prov_ICT	prov_Sindh	prov_KP	prov_Balochistan	cat_perishable	cat_branded_packaged	cat_prepared_food
Islamabad	ICT	perishable	1	0	0	0	1	0	0
Lahore	Punjab	perishable	0	0	0	0	1	0	0
Karachi	Sindh	perishable	0	1	0	0	1	0	0
Peshawar	KP	perishable	0	0	1	0	1	0	0
Quetta	Balochistan	perishable	0	0	0	1	1	0	0
Islamabad	ICT	branded_packaged	1	0	0	0	0	1	0
Islamabad	ICT	prepared_food	1	0	0	0	0	0	1

3. Exploratory data analysis

3.1 Univariate summary

Table 5 reports each variable at its own unit of analysis. City attributes take 17 values, so city-level relationships are always analysed with n = 17, never with the 23,409 rows. **Skewed distributions** show up where the resistant and non-resistant measures disagree. The within-city price range has mean 6.8% but median 1.4% (skewness 3.55). District urban population has mean 3.52 million but median 1.62 million. For city premiums, Larkana's mean (-4.9%) and median (-0.2%) differ by 4.8 points, because a few items drive the mean. We report medians alongside means wherever the two disagree. Weekly price changes are dominated by zeros: 75% of consecutive-week food changes are exactly 0, so their median and IQR are both 0.

Table 5. Summary statistics for key numerical variables: central tendency (mean, median), spread (SD, IQR) and skewness.

variable (unit of analysis; unit)	n	mean	median	sd	iqr	skew	min	max
price_z_item_week (food; z-score, unitless)	14539	-0.000	-0.043	0.970	1.227	0.189	-3.865	3.795
rel_price (food; log points)	14607	-0.000	0.000	0.130	0.110	-0.144	-1.239	0.708
dlog_price (food; log points, consecutive weeks)	8115	-0.003	0.000	0.072	0.000	-1.300	-0.981	0.709
within_city_range_pct (food; %)	14607	6.812	1.395	11.665	9.384	3.549	0.000	152.858
single_quote (food; 1 if MIN = MAX, else 0)	14607	0.486	0.000	0.500	1.000	0.056	0.000	1.000
fies_mod_sev_pct (17 cities; %)	17	15.314	16.380	5.926	8.310	0.647	7.930	29.260
illiteracy10_pct (17 cities; %)	17	35.756	34.000	15.262	20.155	1.080	15.000	78.000
out_of_school_pct (17 cities; %)	17	28.573	25.739	15.495	19.000	0.627	10.000	60.000
no_tap_water_pct (17 cities; %)	17	76.782	81.740	20.667	18.130	-1.121	31.680	100.000
deprivation_composite (17 cities; 0-1)	17	0.427	0.349	0.217	0.261	0.840	0.138	0.974
pop_urban (17 city districts; millions)	17	3.524	1.619	4.940	2.485	2.298	0.048	18.868
log_pop_urban (17 city districts; log persons)	17	14.368	14.298	1.341	1.176	-0.725	10.787	16.753

3.2 City food price premiums

For each city we average `rel_price` over weeks for each food item. The item-level means are the observations, so n is 31-32 items, not thousands of rows. Islamabad's mean premium is +14.1% (95% t-interval [8.9, 19.6]), followed by Rawalpindi +10.2% and Karachi +5.2%. Sukkur (-7.1%) and Bannu (-7.4%) are lowest (Fig. 1). 8 of 17 cities differ from zero at α = 0.05. After a Bonferroni correction for 17 tests, 4 remain: Islamabad, Rawalpindi, Sukkur, Bannu. Premiums persist: of 3,269 consecutive-week pairs in which a city was above the cross-city median, 92.7% stayed above it the next week. Restricting to pairs where the price actually changed, the figure is 84.1%.

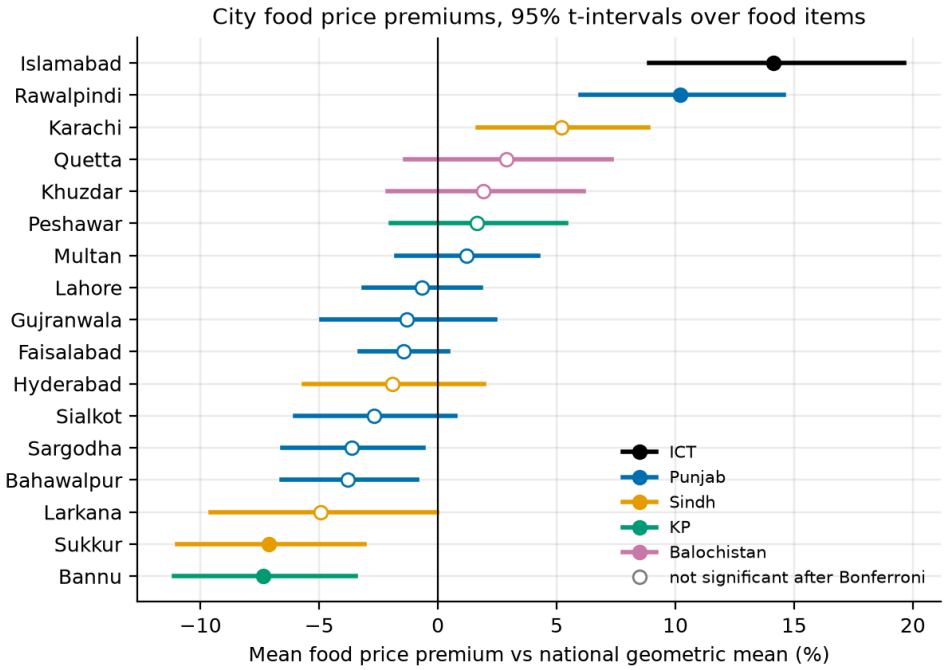


Fig. 1. Mean food price premium of each city relative to the national geometric mean, with 95% t-intervals over food items (n = 31-32). Colour shows province; filled markers remain significant after Bonferroni correction (α = 0.05/17). Islamabad and Rawalpindi pay most; Sukkur and Bannu least.

3.3 Dispersion by food category

The coefficient of variation (CV) puts items priced in different units on one scale. Mean cross-city CV is highest for perishables (0.157, 11 items) and prepared food (0.144), lower for storable staples (0.107, 10 items), and lowest for branded packaged goods (0.026), which carry printed or distributor prices (Fig. 2). A one-tailed Welch comparison of perishables against storables gives p = 0.037, and a rank-based check gives p = 0.057. With about ten items per group this is suggestive, not conclusive. Categories were assigned on storability and pricing mechanism before any dispersion was computed.

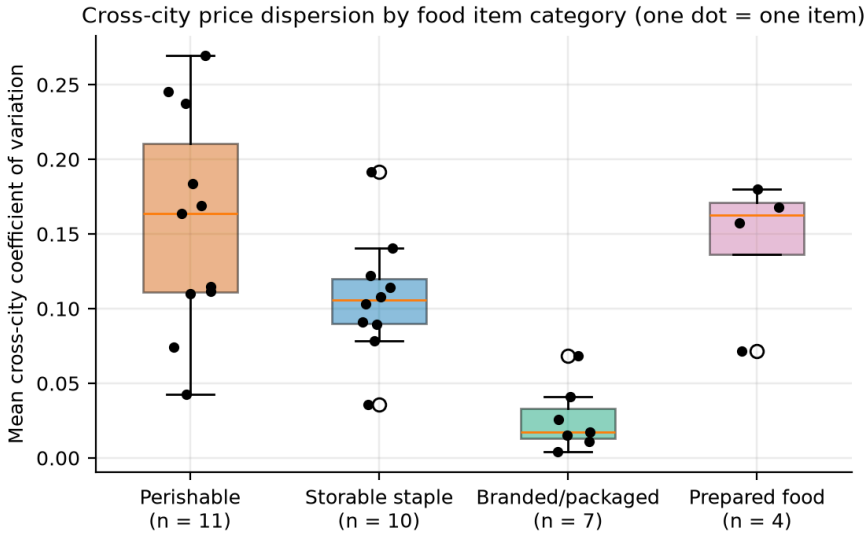


Fig. 2. Mean cross-city coefficient of variation (unitless) of each food item, grouped by category; one dot per item. Perishables are the most dispersed across cities; branded packaged goods barely differ.

3.4 Outlier diagnostic

We apply both rules taught in the course to `rel_price` and to weekly log changes, each at two tiers (Table 6, Fig. 3). **Masking and variance inflation:** the two rules disagree sharply. On `rel_price`, the IQR is narrow because many branded items are priced identically everywhere, so the quartile fences sit close and flag 3,649 values. The SD, by contrast, is inflated by the extreme values themselves, which pulls the 3-SD fence outward and masks them: only 434 are flagged. On weekly changes the IQR is exactly 0, so the quartile rule flags every non-zero change and is unusable. We checked the eight most extreme relative prices against the raw workbooks. 7 equal PBS's own national minimum or maximum for that item-week, and the other is a genuine printed quote (Sukkur tomatoes, PKR 20 per 1 kg). They are therefore true prices: we keep and flag them rather than trim them, because large moves are what H3 studies.

Table 6. Outliers by the quartile rule (beyond Q1 − k·IQR or Q3 + k·IQR) and the z-score rule (|z| > 2, 3). Units: log points.

variable	n	Q1	Q3	IQR	mean	sd	> 1.5 IQR	> 3 IQR	|z| > 2	|z| > 3
rel_price, all 51 items	23326	-0.037	0.046	0.084	0.000	0.133	3649	1251	1598	434
weekly log change, all items	12959	0.000	0.000	0.000	0.001	0.062	2502	2502	493	296

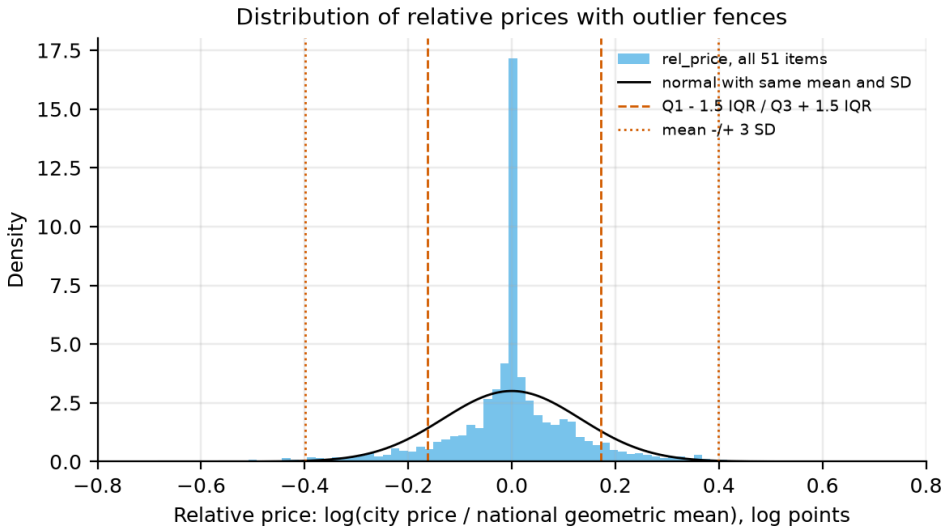


Fig. 3. Distribution of relative prices (log points; all 51 items × 17 cities × 27 weeks) with the 1.5 IQR fences (dashed) and the ±3 SD fences (dotted), plus a normal curve with the same mean and SD. The spike at 0 and the heavy tails explain why the two rules disagree.

3.5 Relationships: premium, deprivation and data quality

Following Anscombe's advice, we plot before computing any coefficient (Fig. 4). The relationship runs **against** the "poor pay more" expectation. The three least deprived cities (Karachi, Islamabad, Rawalpindi) have the highest premiums, while several of the most deprived (Bannu, Sukkur, Larkana) pay below the national reference. The Spearman correlation between premium and the deprivation composite is ρ = -0.56 (p = 0.020, n = 17). Dropping each city in turn keeps ρ between -0.72 and -0.48. Against FIES food insecurity alone, ρ = -0.32 (p = 0.209), which is not significant. The tap-water indicator is a weak deprivation measure: it rates Gujranwala and Sialkot (95% and 94% of households without tap water) as worse off than Quetta or Peshawar, and Karachi and Hyderabad as best off. This mostly reflects reliance on motor pumps and filtration plants in Punjab

cities. Without it, the composite gives ρ = -0.50 (p = 0.041). Khuzdar and Quetta are the exception: their premiums are above zero, though not individually significant, and they have the highest food insecurity in the panel, so burden concentrates there. These are associations across 17 cities, with deprivation data from 2019-20, and support no causal claim.

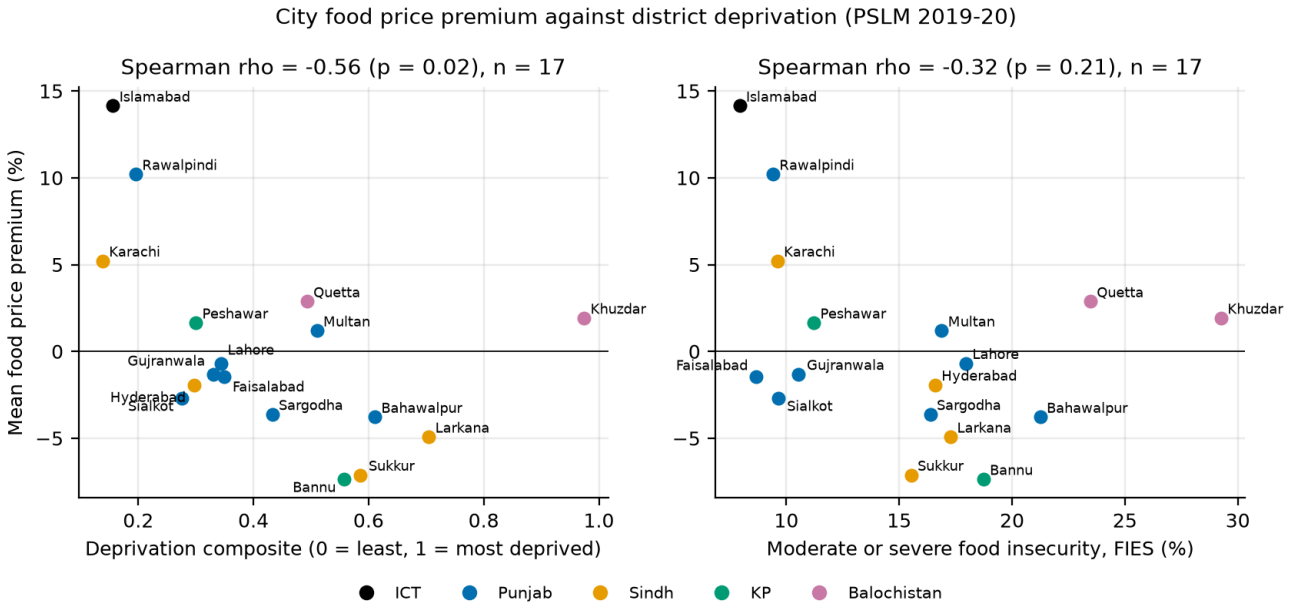


Fig. 4. City mean food price premium (%) against the PSLM 2019-20 deprivation composite (left; 0-1, higher = more deprived) and moderate-or-severe food insecurity (right; % of population). Every city is labelled; colour shows province; n = 17.

Table 7. Spearman correlation matrix of city-level variables (n = 17).

Spearman ρ (n = 17)	premium	deprivation composite	FIES food insecurity	log urban population	share MIN = MAX
premium	1.00	-0.56	-0.32	0.41	-0.74
deprivation composite	-0.56	1.00	0.76	-0.56	0.49
FIES food insecurity	-0.32	0.76	1.00	-0.41	0.14
log urban population	0.41	-0.56	-0.41	1.00	-0.41
share MIN = MAX	-0.74	0.49	0.14	-0.41	1.00

Table 8. Contingency table: SPI cities by province and premium tercile.

province	low premium	mid premium	high premium
ICT	0	0	1
Punjab	3	4	1
Sindh	2	1	1
KP	1	0	1
Balochistan	0	0	2

Data quality. Table 7 also shows that the premium correlates negatively with the share of a city's food records where MIN = MAX (ρ = -0.74). MIN = MAX means every quote PBS collected for that item in that city was identical: either a single quote or several equal ones. The published average then carries no information about price spread within the city. This share ranges from 20% in Karachi to 91% in Sukkur (Fig. 5), a measurement-quality disparity that H4 carries forward. **Representation.** The census districts that contain the 17 SPI cities hold 65% of the urban population of the four provinces and Islamabad. Because each city is smaller than its district, this is an upper bound on what the SPI markets represent. The panel includes no rural markets, AJK or Gilgit-Baltistan, and only two cities each for KP and Balochistan.

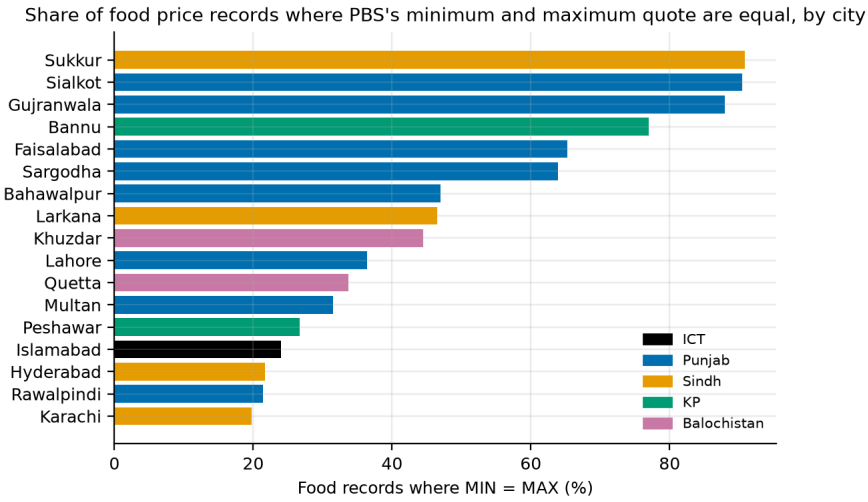


Fig. 5. Share of each city's food price records (%) in which PBS's minimum and maximum quote are equal, coloured by province. High shares mean the published city average carries no information on within-city price spread.

4. Hypotheses and analytical plan

Each hypothesis is mapped to master-dataset columns in Table 9. Targets marked † are derived from existing columns, with the rule for each fixed in Milestone 02. Milestone 03 builds the final H3 label (Section 5) and tests H3 (Sections 6-8); the base rate and row count quoted for H3 below are the whole-period description from Milestone 02, which the models never use.

- **H1, persistent premiums (SDG 2.1).** (a) In $\log p_{c,i,t} = \alpha_{i,t} + \beta_c + \varepsilon$ with $\Sigma\beta_c = 0$: H0: $\beta_c = 0$ for every city c; H1: $\beta_c \neq 0$ for at least one c. (b) H0: P(above median at t+1 | above at t) = 0.5; H1: > 0.5, tested on consecutive weeks and repeated for changed prices only. If some cities persistently pay more for the same basket, access to food differs by place in a way the national figure hides.
- **H2, perishability and dispersion (SDG 2.c).** H0: $\mu_{CV,\ perishable} = \mu_{CV,\ storable}$; H1: $\mu_{CV,\ perishable} > \mu_{CV,\ storable}$. Greater dispersion in perishables points to market-functioning frictions.
- **H3, abnormal next-week movement (SDG 2.c, early warning).** H0: a classifier on lagged features achieves AUC = 0.5 on a time-ordered test set; H1: AUC > 0.5. "Abnormal" means a change larger than the item's own 90th percentile of non-zero changes in the training weeks. We use non-zero changes because 75% of weekly food price changes are zero, and the plain 90th percentile would be 0 for 13 of the 32 food items. Base rate: 2.9%.
- **H4, representation and model fairness (SDG 10).** H0: model error rates and data quality (share of MIN = MAX records) are equal across city groups (province, population tercile, deprivation tercile); H1: at least one group differs.
- **H5, burden versus price (SDG 2.1, SDG 10).** H0: Spearman ρ(city premium, deprivation) = 0; H1: ρ ≠ 0, two-sided, n = 17. EDA already points to ρ < 0 (§3.5). The test will be repeated under both deprivation definitions.

Table 9. Variable mapping: target, predictors and controls for each hypothesis, with column names from the master dataset or its views.

Hypothesis	Target (Y)	Predictors (X)	Controls / confounders	Data	Effective n
H1a premiums	log_price	city (sum-to-zero dummies of city_code)	item × week fixed effects (item_id, week_end)	master, food rows	17 cities; 32 items × 27 weeks
H1b persistence	above_median	above_median_prev	price_changed, item_category; rows with is_consecutive	master, food rows	17 cities (one rate each)
H2 dispersion	cv, averaged per item	item_category (perishable vs storable_staple)	n_quoting	item_week_dispersion view	11 vs 10 items
H3 abnormal move	abnormal_next_week † (final rule in §5)	|dlog_price|, |dlog_price_lag1| with observed flags, price_changed, rel_price, within_city_range_pct, single_quote, cat_* dummies	prov_* dummies, log_pop_urban, deprivation_composite	model_ready_food	7,095 labelled rows
H4 representation	model error † (M03); single_quote (now)	province; terciles of pop_urban and deprivation_composite †	item_category	master / city table	17 cities in 3-5 groups
H5 burden	city premium = city mean of rel_price †	deprivation_composite, fies_mod_sev_pct	log_pop_urban, city share of single_quote	city table (17 rows)	17

Analytical plan: what Milestone 03 did. Milestone 03 is the classification milestone, so it tests H3.

- **Done:** logistic regression, a decision tree and KNN against two naive benchmarks. Validation uses a time-ordered split and forward-chaining cross-validation, because random folds leak later weeks of the same city-item series (§6.4 measures how much). Models are evaluated with ROC-AUC, PR-AUC and the full Unit 10 metrics, not accuracy alone: a model that always predicts "normal" is 97% accurate on the test weeks. Errors are broken down by city group as a descriptive preview of H4 (§8).

- **Moved to Milestone 04:** tree ensembles (beyond the methods covered so far), the formal H4 fairness test, and the fixed-effects estimate of H1. H2 and H5 were tested descriptively in Milestone 02.
- **Still closed:** the post-cutoff holdout. It is opened once, in the Week 14 new-data milestone.

5. Baseline models and justification

5.1 Prediction task, label and features

Unit and target. One row is one city × food item × week *t*. The target `y` = 1 if the item's price in that city moves abnormally between week *t* and the next consecutive release *t*+1. "Abnormal" means |Δlog price| above the item's own 90th percentile of **non-zero** weekly changes, because the plain 90th percentile is 0 for 13 of the 32 food items. Items with fewer than five non-zero changes in the training rows get no threshold and are dropped (5 items in the final split). The positive class is the abnormal move, the event a price monitor wants to catch. **Thresholds are fitted on training rows only**, separately in every fold and in the final split. If they were fitted on all weeks, the test weeks would help define what counts as abnormal.

Why only 15 of 27 weeks carry a label. A label needs the next week's price, and only 15 weeks are followed by a release 6-8 days later. This is a gap in PBS's published weekly archive, not in our method. Described with whole-period thresholds, the labelled data has 7,095 rows (473 per week), of which 205 are abnormal (2.9%). The models never see these whole-period thresholds.

Features. Every feature is known at week *t* (Table 10). A change that cannot be computed because the previous week was not published (40% of rows for this week's change, 73% for last week's) is set to 0 *and* flagged with a `has_` indicator. The model can then tell "no change" from "not observed" (the Milestone 02 rule: flag, don't impute). We deliberately do not add one dummy per item: with about 200 positives this would overfit, and it would swamp KNN's distances.

Table 10. Model features and why each might matter. Reference levels: storable staple (category) and Punjab (province).

feature	why it might matter (all known at week t)
abs_dlog	size of this week's change: prices that just moved may keep moving or snap back
has_dlog	1 if this week's change is observed (previous week published); flag instead of guessing
price_changed	1 if the price changed at all this week; most weeks prices are flat
abs_dlog_lag1	size of last week's change: a longer momentum window
has_lag1	1 if last week's change is observed
rel_price	log price relative to the national geometric mean: a city far from the others may correct
within_city_range_pct	(max - min) / avg across the city's outlets: disagreement may precede a move
single_quote	1 if min = max (one effective quote): a data-quality flag (H4)
cat_perishable	perishable item (reference: storable staple); expected to be most volatile (H2)
cat_branded_packaged	branded / packaged item; printed prices change rarely
cat_prepared_food	prepared food item
prov_Sindh	province dummy (reference: Punjab)
prov_KP	province dummy (reference: Punjab)
prov_Balochistan	province dummy (reference: Punjab)
prov_ICT	Islamabad Capital Territory dummy (reference: Punjab)
log_pop_urban	log urban population of the city's district: market size
deprivation_composite	PSLM 2019-20 district deprivation (0-1): do poorer places see more shocks?

5.2 Benchmarks and models

Table 11 lists the five approaches. The two benchmarks set the floor. The **majority class** shows why accuracy is misleading here. The **category base-rate rule** encodes the domain view that perishables move more, which Milestone 02 supported (H2). It is a tough baseline: it needs no fitting beyond four rates, and any model that cannot beat it is not worth deploying. The three course models represent three different ideas: a linear probability model, axis-aligned rules and similarity. Logistic regression and the tree use `class_weight="balanced"`, which weights each positive row by n/(2 × positives) and each negative by n/(2 × negatives), so the rare class counts as much as the common one. KNN has no class weights; its min-max scaler, like the logistic model's z-score scaler, sits inside an sklearn `Pipeline`, so it is refitted on the training rows of each fold.

Table 11. Naive benchmarks and baseline models, with the reason each was chosen, its tuning grid and the setting selected in cross-validation.

Model	How it works	Why it fits this problem	Grid (tuned in forward-chaining CV)	Chosen
Majority class (benchmark 1)	Always predicts 'normal'	Shows that accuracy misleads on rare events (Unit 10)	none	–
Category base-rate rule (benchmark 2)	Score = item category's abnormal rate in training weeks	One-line domain heuristic: 'perishables move more'. A model must beat it to be worth deploying	none	–
Logistic regression	Linear log-odds; z-scored inputs; balanced class weights	Probabilities for ROC and cost thresholds; stable with few positives; odds ratios an analyst can read	C ∈ {0.01, 0.1, 1, 10}	C=0.01
Decision tree (CART)	Recursive if-then splits; balanced class weights	Rules a price committee can follow without software; captures interactions; no scaling	criterion ∈ {gini, entropy}; depth 2-6; min leaf ∈ {10, 30, 60}	criterion=gini, max_depth=2, min_samples_leaf=60
KNN	Vote of the K nearest rows; min-max scaling fitted on training rows	A different idea: similar city-item situations behave alike; no training step	K ∈ {5, 15, 25, 51, 101, 201}; uniform/distance; Euclidean/Manhattan; all vs continuous features	n_neighbors=201, weights=distance, metric=manhattan, features=all

6. Experimental setup

6.1 Time-ordered split, purge and stratification

We train on the first 11 labelled weeks (2025-07-17 to 2026-03-26) and test on the last 4 (2026-04-02, 2026-06-04, 2026-08-13, 2026-08-20): 73% / 27% by weeks. **Purge rule:** a row at week *t* is labelled from week *t*+1, so a training row whose *t*+1 is an evaluation week is dropped. Here the last training week (2026-03-26) is followed by the first test week, so its 541 candidate rows are purged, and the final models learn from 10 weeks. After purging and labelling, the training set has 4,560 rows (135 abnormal, 2.96%) and the test set has 1,824 rows (54 abnormal, 2.96%): 71% / 29% by rows. The two base rates are nearly equal, so the test weeks are not unusually calm or volatile. The fitted sample (6,384 rows) is smaller than the 7,095-row description in §5.1 for two reasons: the purged week is removed, and thresholds fitted on training rows alone leave 5 items without a label, against 4 under whole-period thresholds.

Why not a random stratified split? The brief's example is a random stratified split. Our rows are a panel: the same 17 × 28 city-item series recur every week, and neighbouring weeks share shocks. A random split would train on later weeks to predict earlier ones, which PBS could never do in practice. It also lets the model learn week-specific shocks that appear on both sides of the split. We keep time order. The purpose of stratification, keeping the rare class represented in every part, is met differently: every validation fold and the test set are checked to contain positives (16-28 per validation week; 54 in test), and logistic regression and the tree use balanced class weights. Class weights change what the model optimises; they are not a substitute for stratification, which is why the positive counts are checked. §6.4 compares random and time-ordered folds.

6.2 Forward-chaining cross-validation

Outer boundary. Cross-validation may only use labels that end before the test period. The last training week (2026-03-26) is labelled from the first test week, so it is excluded from CV as well as from final training. This leaves 10 development weeks (to 2025-12-18). An earlier version of our code used that week as a CV validation week, which let test-period prices shape its labels; an independent review caught this, and every result here comes from the corrected design. The code now asserts that every CV label, in training and validation, ends before the first test week. **Folds.** Within the development weeks we use an expanding window (Fig. 6). Fold 1 trains on labelled weeks 1-5 and validates on week 6. Fold 2 trains on weeks 1-6 and validates on week 7, and so on, giving 5 folds. Each fold rebuilds its own label thresholds, scaler and model from its training weeks and applies the purge rule (4 folds lose their last training week to it). Each validation week holds only 16-28 abnormal moves, so fold-to-fold variance is large. We report the mean ± SD across folds and every fold's AUC (Appendix Table A1).

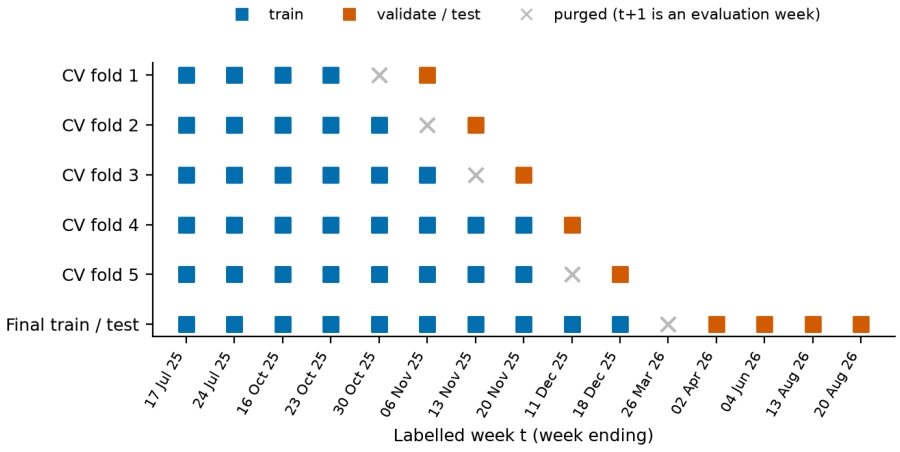


Fig. 6. Forward-chaining folds and the final split, by labelled week t. Blue: training; orange: validation or test; grey ×: training week purged because its t+1 is an evaluation week.

6.3 Hyperparameter tuning

Each grid in Table 11 runs inside the five folds; no test-period information enters this step. Settings are compared by mean validation **ROC-AUC**, which does not depend on a threshold and is not fooled by the majority class. When several settings are within one fold-SD of the best, we take the simplest: smallest C, smallest depth with the largest leaves, or largest K. With so few positives per fold, the difference between near-equal settings is mostly noise (Fig. 7). **Logistic regression:** C hardly matters. C = 0.01, the strongest regularisation, has the best mean (0.641 ± 0.102). **Decision tree:** depth 4 had the best mean (0.673 ± 0.052), but depth 2 is within one SD, so we keep the simpler tree (0.641 ± 0.117); a depth-2 tree is also four readable rules. **KNN:** AUC rises steadily with K and peaks at the grid's edge (n_neighbors=201, weights=distance, metric=manhattan, features=all; 0.657 ± 0.128). Using the four continuous features alone is clearly worse (best 0.607), so the category dummies carry most of what KNN finds.

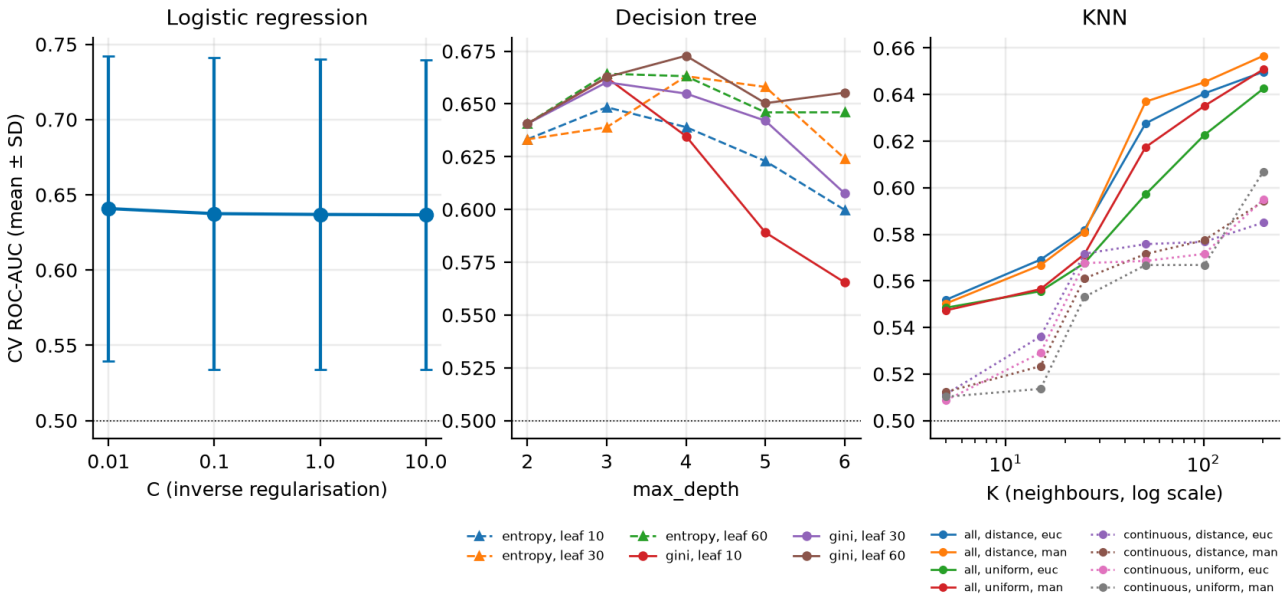


Fig. 7. Tuning results: mean validation ROC-AUC over five forward-chaining folds for each setting (left panel shows ± 1 SD). The dotted line is chance (0.5).

6.4 Leakage checks

Random folds are over-optimistic. For the chosen logistic model, ordinary shuffled `StratifiedKFold` (5 folds) on the training rows gives AUC 0.710 ± 0.038, against 0.641 for forward chaining: a gap of 0.070. This is a comparison, not a controlled measurement of leakage: the two schemes also differ in training size, and the random folds reuse labels fitted on all training rows. But it points the way we expected: ignoring time order makes the model look better than it is. **Shuffled-label sanity check.** Refitting the final logistic model on permuted training labels and scoring the real test labels gives AUC 0.583 for the seed-42 shuffle. Over 200 shuffles the mean is 0.503 (5th-95th percentile 0.374-0.626), and 0 of 200 reached the real test AUC. This is consistent with the pipeline behaving correctly, but it is a sanity check, not proof that nothing leaks (it could not have detected the CV boundary problem above). It is also not a formal test of H3, because rows in the same week are not exchangeable and tuning is not repeated under the null. **Asserts in the code:** no week after the cutoff is read; no evaluation week is in any training set; every CV label ends before the test period; thresholds come from training rows; the purge is applied; the mean shuffled AUC is within 0.03 of 0.5.

7. Results

Table 12 reports the test set: one final evaluation of each model, with settings fixed in cross-validation. The diagnostics in §8 reuse these same test predictions; nothing on the test set was used to choose a model, setting or threshold. Table 13 reports the same metrics as mean ± SD across the five folds. Threshold-based metrics use 0.5. **PR-AUC** (computed as average precision, sklearn's step-wise estimate of the area under the PR curve) matters because, with about 3% positives, ROC-AUC can look respectable while precision stays tiny. A random model's PR-AUC equals the base rate (0.030 on test), so that is the bar to compare against.

Table 12. Test-set metrics (4 weeks, 1,824 rows, 54 abnormal). Threshold 0.5 for accuracy to F1. ROC-AUC with a percentile bootstrap 95% CI (2,000 resamples of test rows, seed 42).

model	accuracy	precision	recall	specificity	F1	ROC-AUC [95% CI]	PR-AUC	flagged at 0.5
Majority class	0.970	0.000	0.000	1.000	0.000	0.500 [0.500, 0.500]	0.030	0
Category base-rate rule	0.970	0.000	0.000	1.000	0.000	0.708 [0.659, 0.753]	0.051	0
Logistic regression	0.570	0.050	0.759	0.564	0.095	0.692 [0.636, 0.746]	0.052	813
Decision tree	0.868	0.035	0.130	0.890	0.055	0.610 [0.576, 0.650]	0.038	201
KNN	0.968	0.000	0.000	0.998	0.000	0.561 [0.487, 0.636]	0.038	4

Table 13. Forward-chaining cross-validation metrics: mean ± SD over 5 validation weeks (threshold 0.5).

model	accuracy	precision	recall	specificity	F1	ROC-AUC	PR-AUC
Majority class	0.946 ± 0.017	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000	0.000 ± 0.000	0.500 ± 0.000	0.054 ± 0.017
Category base-rate rule	0.946 ± 0.017	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000	0.000 ± 0.000	0.617 ± 0.107	0.079 ± 0.042
Logistic regression	0.622 ± 0.051	0.088 ± 0.044	0.628 ± 0.221	0.622 ± 0.061	0.152 ± 0.071	0.641 ± 0.102	0.093 ± 0.032
Decision tree	0.622 ± 0.253	0.066 ± 0.061	0.545 ± 0.384	0.628 ± 0.288	0.116 ± 0.100	0.641 ± 0.117	0.096 ± 0.058
KNN	0.944 ± 0.016	0.000 ± 0.000	0.000 ± 0.000	0.998 ± 0.001	0.000 ± 0.000	0.657 ± 0.128	0.107 ± 0.061

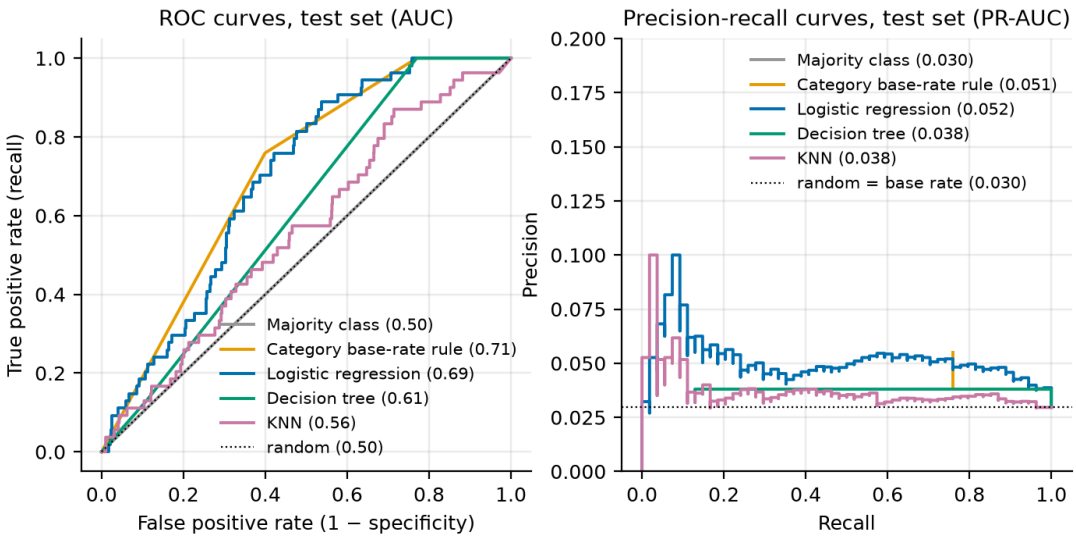


Fig. 8. ROC curves (left; AUC in brackets) and precision-recall curves (right; PR-AUC in brackets; y-axis cut at 0.2) on the test set. The dotted lines are chance: the diagonal, and the base rate.

H3 decision. For logistic regression and the decision tree, the lower bound of the row-level bootstrap 95% CI for test ROC-AUC is above 0.5 (logistic 0.692 [0.636, 0.746]; tree 0.610 [0.576, 0.650]); for KNN it is not (0.561 [0.487, 0.636]). The bootstrap treats rows as independent, but rows in the same week share shocks, so these intervals are too narrow. As a check that respects weeks, we score each test week separately (Appendix Table A3): logistic regression is above 0.5 in all 4 weeks (range 0.542-0.748), the category rule in 3. We therefore **reject H0 for logistic regression in these test weeks**: information available at week *t* ranked next-week abnormal moves better than chance. With only four weeks this is evidence of a modest signal, not an established general result; the Week 14 holdout is the next test. Test PR-AUC is 0.052, under twice the base rate.

Does any ML model beat the simple rule? In cross-validation, logistic regression is ahead of the category rule on average (0.641 ± 0.102 vs 0.617 ± 0.107), but only in 3 of 5 folds. On the test weeks, the category rule is slightly ahead (0.708 vs 0.692). The paired bootstrap difference (logistic minus rule) is -0.015, 95% CI [-0.074, 0.041], so we cannot tell them apart. The depth-2 tree scores below the rule on test (row-bootstrap difference CI [-0.154, -0.036], with the same independence caveat). KNN had the highest CV mean but also the largest SD, and it fell to 0.561 on test. **The honest reading is that most of the usable signal is "what kind of food is it".** The ablation agrees. Logistic regression with only the three category dummies scores 0.617 in CV and 0.708 on test. The full feature set adds about 0.02 AUC in CV and nothing on test.

Accuracy misleads. The majority class is 97.0% accurate and catches nothing. The category rule never reaches a score of 0.5 and KNN does so for only 4 rows, so at that threshold they flag (almost) nothing, and their accuracy-to-F1 rows look like the majority class. Balanced class weights push logistic regression the other way: at 0.5 it flags 813 rows, with recall 76% but precision 5.0%. 0.5 is an arbitrary cut-off. §8 chooses thresholds from costs.

Why KNN struggles (Unit 11 limitations). With about 3% positives, most neighbourhoods of 5-25 rows contain no abnormal move at all, so the vote is 0 almost everywhere. Binary dummies count as much in the distance as real price changes, and irrelevant features add noise. Many rows are identical (no change, not observed), which produces ties. Only very large neighbourhoods give a usable score, and then KNN is a smoothed local base rate, which explains its peak at the edge of the K grid.

Label sensitivity. Under the Milestone 02 "contract" rule (plain 90th percentile; any change counts for 13 items) the test base rate is 7.2%. Logistic regression reaches 0.585 and the category rule 0.523, so the task is harder. The refined label is the one we defend.

8. Asymmetric error costs and confusion-matrix diagnostics

8.1 Who bears each error

The alert is meant for a PBS analyst, or a district Price Control Committee or price magistrate deciding where to send inspectors or which items to watch next week.

- A **false negative** (missed abnormal move) is, at its most harmful, a jump in tomatoes, onions, flour or ghee that nobody was warned about (abnormal moves also include sharp falls, which matter to producers and traders but less to consumers). No early stock release, inspection or public notice happens, and households absorb the shock. Households in more deprived districts spend a larger share of income on food, so the harm falls hardest on those least able to absorb it (H5).
- A **false positive** (false alarm) is an inspection visit or watch-list entry that was not needed. It costs staff time and may cause friction with traders. Repeated false alarms cause *alert fatigue*: if most alerts are wrong, people stop trusting the system.

Cost matrix (Provost and Fawcett, Ch. 7). cost(TP) = cost(TN) = 0, cost(FP) = 1 unit and cost(FN) = r units. We put the cost in one cell only, to avoid the double counting Provost and Fawcett warn about. We do not also credit a "benefit" for true positives. Flagging a row with probability p of being abnormal is worth it when $p \cdot r > 1 - p$, i.e. $p > 1/(1 + r)$. For an average row, p is the training base rate (2.96%), so "flag everything" beats "flag nothing" exactly when $r > (1 - p)/p = 32.8$, close to the inverse base rate (33.8). ***r* is a value judgment made by people, not something the data decides**, so we vary it over {1, 5, 10, 20, 35}. For each r and model, the threshold that minimises total cost is chosen on the pooled out-of-fold CV predictions (all from development weeks) and then applied to the test predictions.

Table 14. Test cost per 1,000 city-item-weeks (in false-alarm units) and, in brackets, the share of abnormal moves caught, for each cost ratio r. Thresholds were chosen on CV predictions only.

policy	r = 1	r = 5	r = 10	r = 20	r = 35
Category base-rate rule	30 (0%)	148 (0%)	296 (0%)	592 (0%)	637 (76%)
Logistic regression	30 (0%)	148 (0%)	430 (46%)	592 (83%)	709 (94%)
Decision tree	30 (0%)	148 (0%)	364 (13%)	747 (100%)	805 (100%)
KNN	30 (0%)	148 (0%)	418 (20%)	711 (48%)	932 (69%)
Flag nothing (= majority class)	30 (0%)	148 (0%)	296 (0%)	592 (0%)	1036 (0%)
Flag everything	970 (100%)	970 (100%)	970 (100%)	970 (100%)	970 (100%)

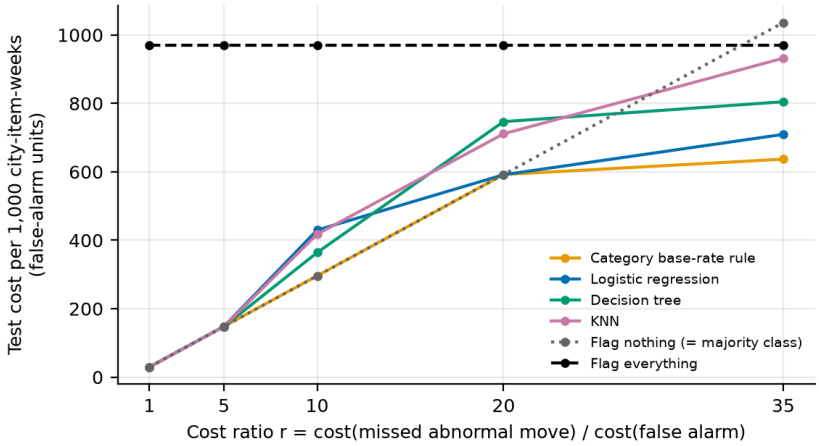


Fig. 9. Test cost per 1,000 city-item-weeks against the cost ratio r. For r ≤ 10 the cheapest policy is to issue no alerts; at r = 35 the category rule is cheapest.

Reading the sensitivity (test weeks only). For $r \leq 10$ the cheapest policy is to flag nothing. At $r = 10$ the three ML models' CV thresholds raise some alerts, and on test each costs more than staying silent. At $r = 20$, logistic regression (591.6) and flagging nothing (592.1) are essentially tied. At $r = 35$ the **category rule** is cheapest (637), ahead of logistic regression (709), flagging everything (970) and flagging nothing (1,036). So, in these test weeks, **no ML model beats the simple rule on cost, and alerting only pays off at all when a missed shock is judged to be worth roughly 20 or more unnecessary inspections.** These are the points we chose to try, not a measured break-even, and the recommended threshold depends on r. We use $r = 20$ below as an illustrative reference (a value judgment, not chosen from the data). At that threshold logistic regression flags 52% of test rows, catches 83% of abnormal moves, and has precision 4.8% (Appendix Table A2 has all models).

Alert budget. A decision-maker may think in terms of capacity rather than r: "we can check N city-items a week". Table 15 ranks each test week's 456 rows by score and takes the top N. With 50 checks a week, logistic regression puts 11 of the 54 abnormal moves in front of an inspector (20%); the category rule, with random tie-breaking, puts 11.0 in front of them. Precision@N stays between 4% and 7%, against 3.0% for random checks.

Table 15. Alert budget, pooled over the four test weeks: share of abnormal moves caught (recall) and share of checks that find one (precision) when only the top N rows per week are checked. Ties (category rule) are counted at their expected value. Per-week results are in Appendix Table A4.

model	top 10/week: recall	top 10/week: precision	top 25/week: recall	top 25/week: precision	top 50/week: recall	top 50/week: precision
Category base-rate rule	0.041	0.055	0.102	0.055	0.203	0.055
Decision tree	0.026	0.035	0.065	0.035	0.130	0.035
KNN	0.037	0.050	0.111	0.060	0.148	0.040
Logistic regression	0.037	0.050	0.130	0.070	0.204	0.055

Equity weighting (preview of the Milestone 04 fairness audit). If a missed shock counts 1, 1.5 or 2 times as much in low-, mid- and high-deprivation cities, the cost-minimising choices shift towards flagging. At $r = 10$ the logistic threshold falls to 0.47 and recall rises to 83%, although on test flagging nothing is still cheapest. At $r = 35$ logistic regression becomes the cheapest policy. Weighting by need changes who gets protected, and that is a decision for policy-makers, not for us.

8.2 Confusion matrices

Fig. 10 uses the Unit 10 layout: actual class in rows, predicted in columns, positive (abnormal) class first, with counts and row percentages. So the top-left cell over its row is recall, and the bottom-right over its row is specificity. The majority class and the category rule (at $r = 20$) have an empty first column. Logistic regression at 0.5 catches 41 of 54 abnormal moves at the price of

772 false alarms. The $r = 20$ threshold trades a little more recall for more false alarms. The depth-2 tree at its $r = 20$ threshold flags every non-branded item, which is the category rule in disguise.

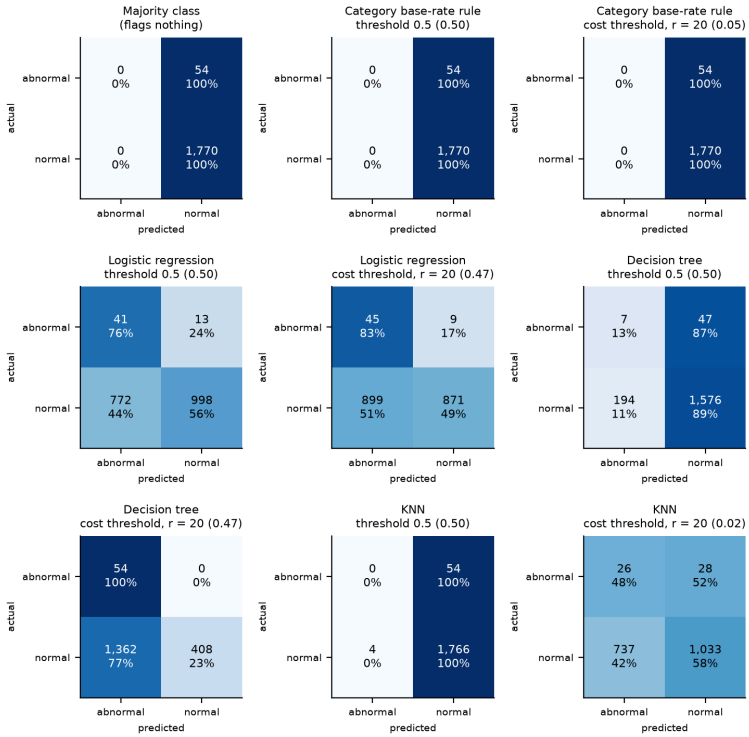


Fig. 10. Test-set confusion matrices: counts and row percentages. Rows are the actual class, columns the predicted class, abnormal first. "cost threshold" is the threshold minimising cost at r = 20 on CV predictions; its value is in brackets.

8.3 Where the errors fall, and edge cases

Table 16 breaks down the logistic model's errors at the $r = 20$ threshold. This is descriptive only; the formal test is H4 in Milestone 04, and many cells hold under 20 positives. Two patterns stand out.

- **Flags concentrate in perishables.** The FP rate is 84% for perishables against 0% for branded goods.
- **Flags concentrate in high-deprivation cities.** The FP rate is 69% there against 38% in low-deprivation cities. The miss rate runs the other way: 10% in high- and 31% in low-deprivation cities.

By province, Islamabad's 2 abnormal moves were all missed, while Sindh's 10 were all caught. Both groups are tiny.

Table 16. Logistic regression errors on the test set at the r = 20 threshold, by group. FN rate = share of abnormal moves missed; FP rate = share of normal rows flagged. Terciles are over the 17 cities.

grouping	group	rows	positives	base rate	flagged	FN rate	FP rate
item category	branded_packaged	408	0	0.000	0	–	0.000
item category	perishable	748	41	0.055	632	0.122	0.843
item category	storable_staple	668	13	0.019	312	0.308	0.463
city-size tercile	large	640	18	0.028	278	0.056	0.420
city-size tercile	medium	540	20	0.037	284	0.300	0.519
city-size tercile	small	644	16	0.025	382	0.125	0.586
deprivation tercile	high	648	20	0.031	449	0.100	0.686
deprivation tercile	low	644	16	0.025	248	0.312	0.377
deprivation tercile	mid	532	18	0.034	247	0.111	0.449
test week	2026-04-02	456	18	0.039	207	0.167	0.438
test week	2026-06-04	456	18	0.039	266	0.111	0.571
test week	2026-08-13	456	9	0.020	262	0.111	0.568
test week	2026-08-20	456	9	0.020	209	0.333	0.454

Edge cases (Table 17; the full ten of each are in the notebook).

- **Every missed abnormal move had a flat or unobserved price at week *t*** (100% of the 9 false negatives have `price_changed` = 0). Moves that come with no visible recent history are invisible to any model built on recent changes.
- **The most confident false alarms** are perishables in Sukkur, Bannu and Larkana. Most are records where PBS's minimum and maximum quotes are equal (one effective quote), which changed this week and then did not move again. A single quote that jumps may reflect one shop rather than the market. That links model errors to the measurement-quality concern in H4.
- **Timing.** The two earlier test weeks have about twice the base rate of the August weeks (4.0% and 4.0% against 2.0% and 2.0%). The 2 April week comes about 12 days after Eid-ul-Fitr, observed in Pakistan on 21 March 2026, and the 4 June week about a week after Eid-ul-Adha, observed on 27 May 2026 (APP, 2026; Radio Pakistan, 2026). Post-festival price adjustment

is *one possible explanation that we have not tested*; the dates alone are not evidence of a holiday effect, and some of these abnormal moves are falls rather than rises.

Table 17. The five most confident false negatives (lowest score among missed abnormal moves) and false positives (highest score among normal rows), logistic regression, r = 20 threshold. Changes are in log points.

error	city	item	week	score	next-week change	threshold	|change| at t	changed at t	MIN = MAX
FN	Quetta	Rice basmati broken	2026-08-20	0.272	-0.072	0.045	0.000	0	0
FN	Quetta	Rice IRRI-6/9	2026-08-20	0.278	-0.061	0.058	0.000	0	0
FN	Multan	Garlic	2026-04-02	0.328	-0.219	0.149	0.000	0	0
FN	Islamabad	Eggs (farm)	2026-06-04	0.377	-0.148	0.057	0.000	0	0
FN	Islamabad	Eggs (farm)	2026-08-13	0.378	-0.096	0.057	0.000	0	0
FP	Larkana	Onions	2026-08-20	0.854	-0.161	0.416	0.379	1	1
FP	Sukkur	Tomatoes	2026-04-02	0.844	0.000	0.624	0.061	1	1
FP	Sukkur	Potatoes	2026-04-02	0.843	0.000	0.187	0.182	1	1
FP	Sukkur	Tomatoes	2026-08-20	0.841	-0.288	0.624	0.223	1	1
FP	Bannu	Eggs (farm)	2026-08-20	0.837	0.000	0.057	0.077	1	1

9. Drivers and initial policy insights

Logistic regression. Inputs are z-scored inside the pipeline, so each coefficient is the change in log-odds per one standard deviation, and e^{coef} is the odds ratio (OR) per SD (Table 18, Fig. 11). For a 0/1 feature, one SD is a fixed fraction of the switch from 0 to 1, so these ORs rank features rather than describe a full category change. They are associations, not causes. The top five:

- `cat_branded_packaged` (OR 0.60): one SD more of being a branded/packaged item is associated with 40% lower odds of an abnormal move next week.
- `price_changed` (OR 1.47): one SD more of a price change this week is associated with about 1.47 times the odds of an abnormal move next week.
- `cat_perishable` (OR 1.33): one SD more of being a perishable is associated with about 1.33 times the odds of an abnormal move next week.
- `has_dlog` (OR 0.79): one SD more of having an observed change this week (previous week published) is associated with 21% lower odds of an abnormal move next week.
- `prov_ICT` (OR 0.80): one SD more of being in Islamabad is associated with 20% lower odds of an abnormal move next week.

`abs_dlog` gets a small coefficient because it is correlated with `price_changed` (r = 0.50 in the training rows), and the two share the credit (Unit 09). The `has_` flags are negative mostly because they mark *which weeks* had a published previous week, so they partly stand in for time period, not for a market effect.

Decision tree. The depth-2 tree splits first on branded/packaged, then on this week's change for other items, and on relative price for branded ones; the diagram is in the notebook (Part D.2) and `reports/m03/figures/fig21_tree.png`. Its impurity importance is 0.58 for `cat_branded_packaged`, 0.29 for `abs_dlog` and 0.12 for `rel_price`. Its leaves read as rules (Table 19). The top rule is: **IF the item is not branded/packaged AND its price moved by more than about 1% this week, THEN the training abnormal rate is 6.9%**, against 3.0% overall. Branded items whose price is at or above the national reference had no abnormal move in training. **Worked Gini (root).** Unweighted, the root holds 135 positives in 4,560 rows (Gini 0.0575); the split sends 130/3,540 left and 5/1,020 right, gain 0.0004. With balanced weights (positive 16.9, negative 0.515), the root is exactly 50/50 (Gini 0.50), the children have Gini 0.494 and 0.239, and the weighted gain is 0.040, the largest available.

Cross-model check. Both models put the branded/packaged flag first, and both give recent price movement a top role (`price_changed` in the logistic model, `abs_dlog` in the tree: two views of the same signal). These are the trustworthy drivers. Perishability, smaller markets and higher district deprivation are positive in the logistic model only. A depth-2 tree can use just three features, so its silence on them is not evidence against them.

Table 18. Top 10 features by \|logistic coefficient\| and their tree importance (blank = unused by the tree).

feature	coef per SD	odds ratio per SD	rank (logistic)	tree importance	rank (tree)
cat_branded_packaged	-0.513	0.599	1	0.584	1.000
price_changed	0.384	1.467	2	0.000	–
cat_perishable	0.284	1.328	3	0.000	–
has_dlog	-0.234	0.791	4	0.000	–
prov_ICT	-0.221	0.802	5	0.000	–
has_lag1	-0.186	0.831	6	0.000	–
log_pop_urban	-0.181	0.834	7	0.000	–
deprivation_composite	0.138	1.148	8	0.000	–
prov_Balochistan	-0.120	0.887	9	0.000	–
within_city_range_pct	-0.106	0.900	10	0.000	–

Table 19. Decision-tree leaves as IF-THEN rules, with their actual abnormal rate on the training rows.

rule	training rows	abnormal	abnormal rate
IF cat_branded_packaged <= 0.5 AND abs_dlog > 0.0101	798	55	0.069
IF cat_branded_packaged <= 0.5 AND abs_dlog <= 0.0101	2742	75	0.027
IF cat_branded_packaged > 0.5 AND rel_price <= -0.00741	287	5	0.017
IF cat_branded_packaged > 0.5 AND rel_price > -0.00741	733	0	0.000

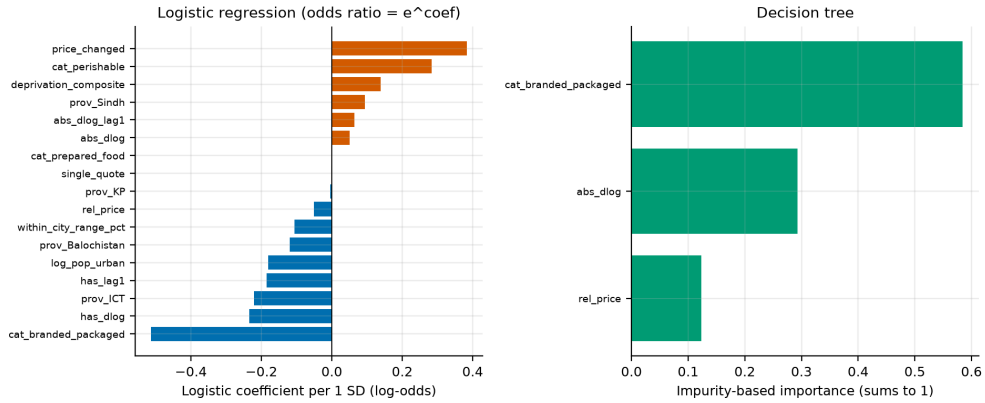


Fig. 11. Feature importance. Left: logistic regression coefficients per 1 SD (red raises the odds of an abnormal move next week, blue lowers them). Right: decision-tree impurity importance.

Initial policy takeaways (SDG 2.c). These are hedged: they rest on four test weeks and modest signal.

1. **Where to look.** Concentrate weekly monitoring on perishables, especially those whose price already moved this week. Branded and packaged goods almost never move abnormally.
2. **At most a triage aid, not a trigger.** With test AUC around 0.7 and precision around 5%, most alerts will be false. A model score can rank where an inspector looks first; it should not trigger stock releases or penalties by itself.
3. **A simple rule already captures most of the signal.** "Watch perishables" matched the logistic model on the test weeks. Any data system PBS or a province builds should be benchmarked against that rule before it is trusted.
4. **Publish every week.** 40% of labelled rows have no previous consecutive week, because the published weekly archive has gaps. The moves we miss are exactly those with no visible recent history. Consistent publication of the weekly annexure is a cheap way to improve early warning.
5. **The threshold is a policy choice.** Whether alerts are worth issuing depends on how many false alarms one missed shock is worth. Below about 20, silence is cheaper. That judgment, and whether to weight misses in deprived districts more, belongs to decision-makers.

10. Limitations and Milestone 04 roadmap

Limitations.

- **Sample size.** Only 205 positives over 15 labelled weeks. The test set has 54 positives in four weeks, so every test number has a wide interval.
- **Intervals and the archive.** The bootstrap treats rows as independent, but rows in the same week share shocks, so the true uncertainty is larger; the per-week check uses only four weeks. Several diagnostics (costs, alert budget, error breakdown) reuse the same test predictions. PBS archive gaps cost 40% of rows their recent history and limit how many weeks can be labelled.
- **Signal.** Class imbalance and modest signal. KNN is unsuitable for this data. Two tuned settings sit at the edge of their grids (C = 0.01, K = 201), which suggests the data favour very smooth models.
- **Label.** The 90th-percentile threshold is a convention. A stricter or looser cut-off would change the base rate and the results (§7).
- **Context variables.** City-level variables (population, deprivation from 2019-20) are constant within a city. Islamabad and Rawalpindi function as one market but are scored separately.
- **Scope.** 17 urban SPI cities only, so possible place-based bias in who gets missed (§8.3). The Milestone 02 limitations on coverage, deprivation data and measurement still apply.

Milestone 04 plan.

- **Advanced models:** random forest and gradient boosting, compared against today's two benchmarks and three baselines under the same forward-chaining CV, purge and cost analysis.
- **Fairness audit (H4):** recall, FN rate and calibration compared across province, population tercile and deprivation tercile, with a hypothesis test, and including whether the single-quote data-quality flag drives errors.
- **Thresholds and ethics:** group-aware thresholds that equalise miss rates, and their cost. An ethical evaluation of who bears the false negatives and the false alarms.
- **Week 14 new data:** the design already committed in Milestones 01-02. First, score the frozen model on the post-cutoff holdout weeks without refitting. Then retrain on the earlier part of the new data and test on a later, unseen part.

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Appendix

Table A1 (part 1). Forward-chaining folds: weeks, purged rows, and rows and positives after labelling. 'items w/o label' = items with fewer than five non-zero changes in that fold's training rows.

fold	training weeks (t)	validation week	purged	train rows	train pos.	val. rows	val. pos.	items w/o label
1	2025-07-17 to 2025-10-30	2025-11-06	541	1416	56	354	28	11
2	2025-07-17 to 2025-11-06	2025-11-13	541	1855	70	371	18	10
3	2025-07-17 to 2025-11-13	2025-11-20	541	2328	87	388	16	9
4	2025-07-17 to 2025-11-20	2025-12-11	0	3240	111	405	26	8
5	2025-07-17 to 2025-12-11	2025-12-18	541	3240	111	405	16	8

Table A1 (part 2). Validation ROC-AUC per fold for the chosen settings.

model	fold 1	fold 2	fold 3	fold 4	fold 5
Majority class	0.500	0.500	0.500	0.500	0.500
Category base-rate rule	0.740	0.666	0.448	0.609	0.624
Logistic regression	0.724	0.758	0.510	0.583	0.630
Decision tree	0.765	0.742	0.485	0.568	0.643
KNN	0.730	0.807	0.466	0.624	0.657

Table A2. Test metrics at the cost-based threshold for r = 20 (chosen on pooled CV predictions, applied once to test).

model	threshold (from CV)	accuracy	precision	recall	specificity	f1	tp	fp	fn	tn
Category base-rate rule	0.053	0.970	0.000	0.000	1.000	0.000	0	0	54	1770
Logistic regression	0.467	0.502	0.048	0.833	0.492	0.090	45	899	9	871
Decision tree	0.471	0.253	0.038	1.000	0.231	0.073	54	1362	0	408
KNN	0.022	0.581	0.034	0.481	0.584	0.064	26	737	28	1033

Table A3. Test ROC-AUC computed separately for each test week (the dependence check used in §7).

week	rows	positives	category rule	logistic	tree	KNN
2026-04-02	456	18	0.740	0.748	0.648	0.705
2026-06-04	456	18	0.785	0.736	0.616	0.559
2026-08-13	456	9	0.757	0.694	0.614	0.455
2026-08-20	456	9	0.449	0.542	0.512	0.406

Table A4. Alert budget per test week: abnormal moves caught (expected count under random tie-breaking) by the category rule and logistic regression. Abnormal moves per week: 2026-04-02 18, 2026-06-04 18, 2026-08-13 9, 2026-08-20 9.

model	checks per week	2026-04-02	2026-06-04	2026-08-13	2026-08-20
Category base-rate rule	10	0.8	0.9	0.4	0.1
Category base-rate rule	25	2.0	2.3	1.1	0.1
Category base-rate rule	50	4.0	4.5	2.1	0.3
Logistic regression	10	0.0	1.0	1.0	0.0
Logistic regression	25	4.0	2.0	1.0	0.0
Logistic regression	50	5.0	4.0	2.0	0.0